

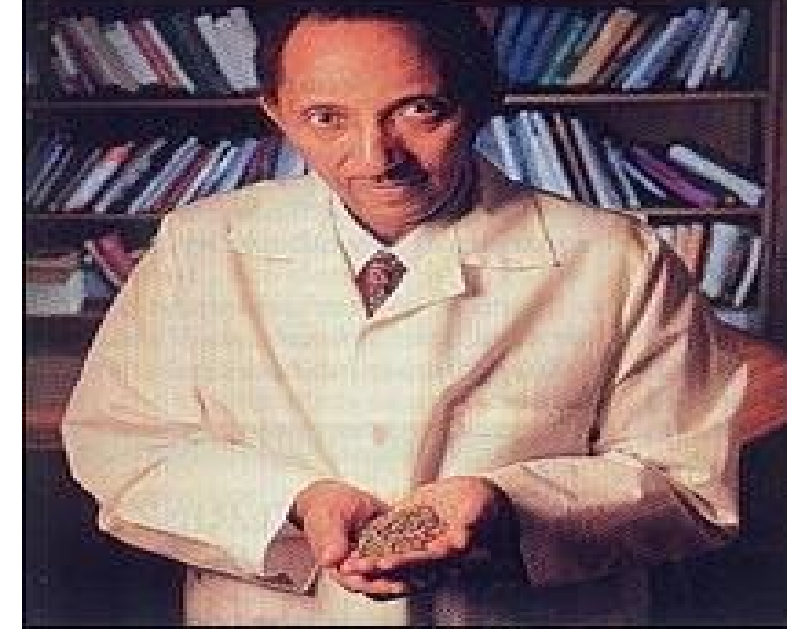
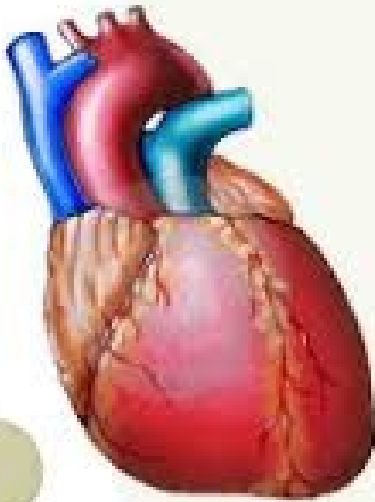
Glycosides

Outline

- Definition
- Distribution & importance
- General properties of glycosides & extraction
- Classification of glycosides

Objective

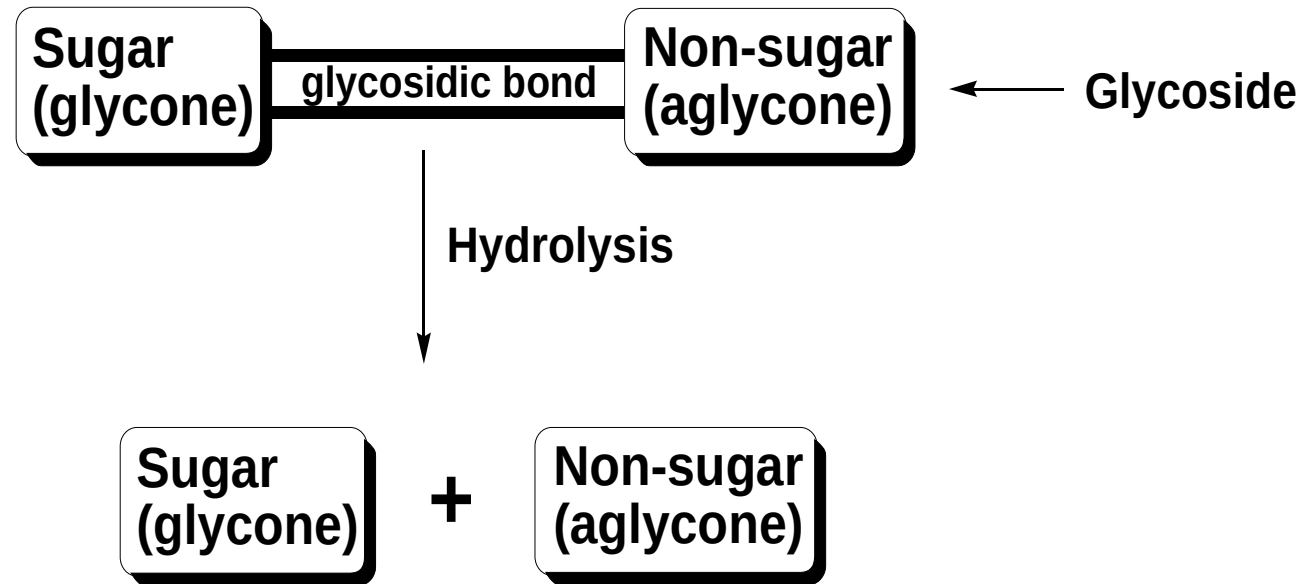
- Define glycosides
- Benefits of glycoside
- Classify and give examples to the different glycosides
- General and specific properties of glycosides



Glycosides

- Definition
 - Is a non-reducing organic compound, usually of plant origin, that is composed of two chemically and functionally independent moieties.
 - The sugar portion is called glycone, while the non-sugar portion is called aglycone or genin
 - post modification of the secondary metabolites catalyzed by plant enzymes like glycosyltransferases
 - Glycosides have long been used as laxatives, cardiostimulant, analgesic, local irritant,

Glycosides Cont...



- Pharmacological activity is due to the aglycone
- The sugar part influences pharmacokinetic

Distribution and Function

Distribution

- Plants, Microorganisms, Animals

Function in plants or organism of origin

- Sanitary or protective activities
- Increase solubility of compounds
- Stabilize some unstable compounds
- Regulation of growth and development
- Serve as sugar reserves
- Serve as pollinator attractants
- Important role in signaling
- An allelopathy (biological phenomenon where one plant inhibits the growth of another)

Extraction and Isolation

- Because of the wide range of physical and chemical properties of glycosides and other constituents associated with them
 - no common general method for their isolation is recommended
- Water, methanol, water-ethanol and ethanol are the most common solvents for extraction of glycosides

Precautions Before Extraction

Deactivation of enzymes:

- Drying for 15-30 min at 100 °C followed by slow drying at a low temperature
- Dipping the fresh material into boiling water or boiling alcohol for 10-20 min
- Boiling the fresh plant material with acetone
- Carrying out the extraction at very low temp
- Freeze-drying of the plant material before extraction (lyophilization)
- Carrying the extraction in the presence of $(\text{NH}_4)_2\text{SO}_4$

Precautions Before Extraction

Maintenance of neutral conditions:

- Neutral pH should be assured before and during extraction because:
Acidity may result in hydrolysis.
- This is overcome by addition of CaCO_3
- Mild alkalinity may sometimes produce racemization

Defatting of fat-rich organs (e.g. seeds) before extraction:

- High amounts of lipoids hinder glycoside extraction
- usually carried with petroleum ether

Chemical Properties

- Occur as a combination or in free form
- D-glucose is the most common sugar
 - More than one sugar may be present
- They are non-reducing
- Most natural glycosides have the β -form
- Most glycosides are hydrolysable

Physical Properties

- Most are crystalline
- Form levorotatory solutions in water and alcohol
- Most have bitter taste
- Solubility

	Water	Alco.	Hyd. Alco.	Organic solvents
Glycosides	□	□	□	□
Aglycones	□	□	□	□

Classification:

- **Atom from the aglycone involved in the glycosidic linkage:**

- Aglycone- O- Sugar O-glycosides
- Aglycone- C- Sugar C-glycosides
- Aglycone- S- Sugar S-glycosides (thioglycoside)
- Aglycone- N- Sugar N-glycosides (glycosylamine)

- **Number of sugars:**

- One sugar Monosides e.g. Salicin.
- Two sugar Biosides e.g. Diosmin.
- Three sugars Triosides e.g. Digoxin.

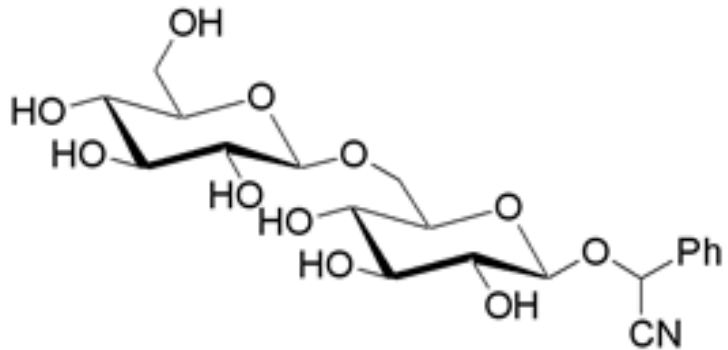
- **Nature of the glycoside:**

- Primary glycosides: Originally present in the plant e.g. Purpurea A
- Secondary glycosides: Resulted from removal of one sugar from the primary glycosides e.g. Digitoxin

- **Type of the glycosidic linkage (stereochemistry):**
 - α - glycosides
 - β - glycosides
- **Source:**
 - Digitalis glycosides
 - Senna glycosides.
- **Therapeutic use:**
 - Analgesic glycosides.
 - Purgative glycosides.
 - Cardiac glycosides.
- **Chemical nature of the aglycone:**
 - Flavone glycosides.
 - Steroidal glycosides.
 - Aldehydic glycosides.
- **Chemical nature of the sugar;**
 - Glucose eg; glucosides
 - Rhamnose eg; rhamnoside

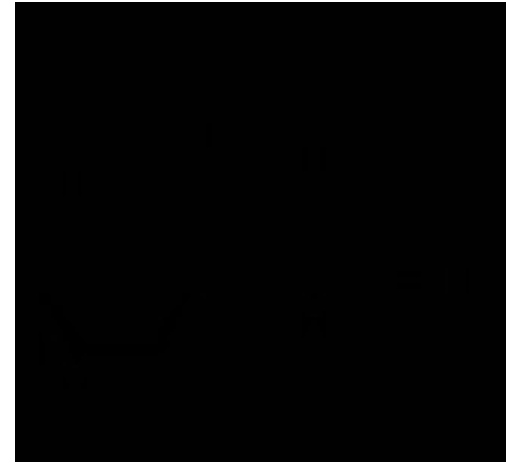
Glycosidic Linkage

O-glycosides



- Are the most abundant glycosides
- Easily be hydrolyzed by boiling in mineral acids

S-glycosides

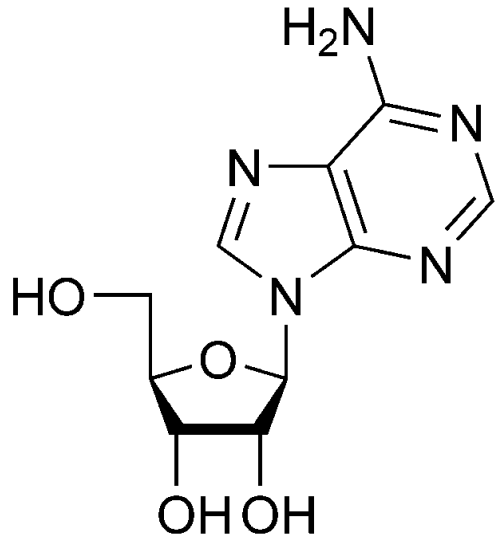


- Natural S-glycosides are very rare
- Easily be hydrolyzed by boiling in mineral acids

Glycosidic Linkage

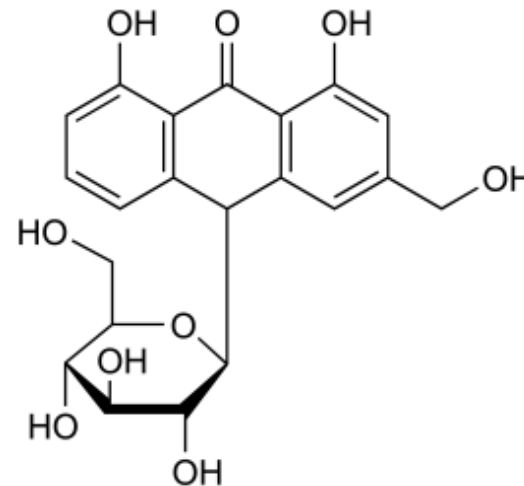
N-glycosides

- easily be hydrolyzed by boiling in mineral acids



C-glycosides

- The sugar is condensed directly to a carbon atom
- are highly resistant to acid hydrolysis

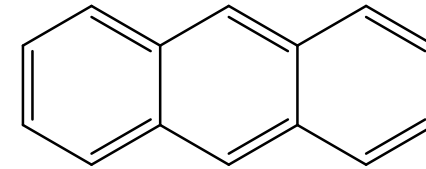


Medicinally and Pharmaceutically Important Glycosides

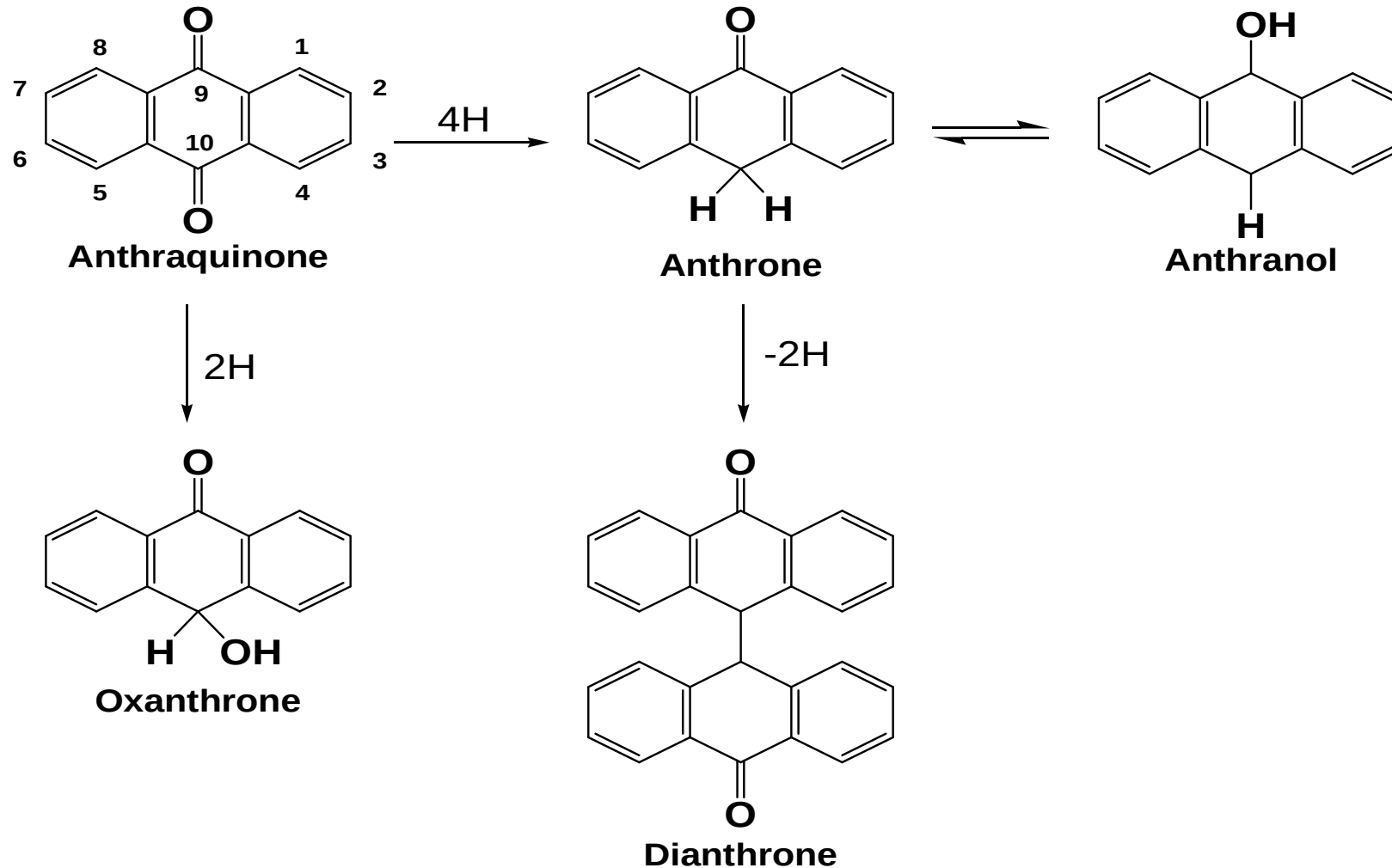


1. ANTHRAQUINONE GLYCOSIDES

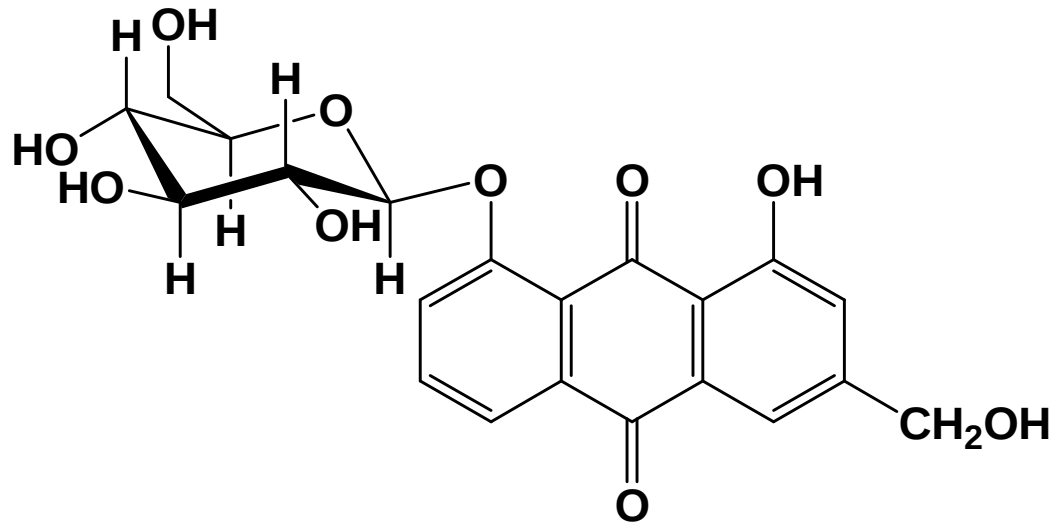
- Anthracene derivative, where three benzene rings are joined together
- Used as Purgatives and dyestuff



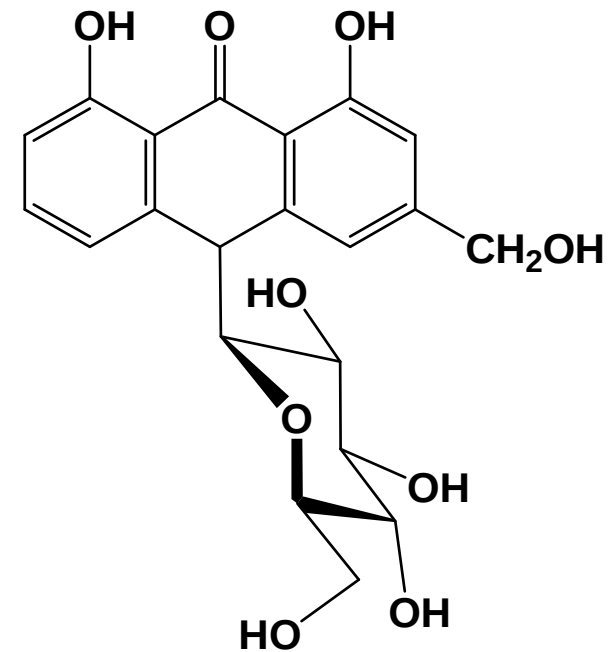
Different derivatives of anthraquinones



Examples



Aloe-emodin 8-glucoside
(an O-glycoside)



Barbaloin
(a C-glycoside)

- **Historically:** Rhubarb, Senna, Aloes and Cascara were all used as purgative drugs
- **Monocotyledons:** Only Liliaceae
 - Most commonly C-glycoside: barbaloin
- **Dicotyledons:** Rubiaceae, Leguminosae, Polygonaceae, Rhamnaceae, Ericaceae, Euphorbiaceae, Lythraceae, Saxifragaceae, Scrophulariaceae and Verbenaceae
 - Also in certain fungi and lichen

- used as laxatives or cathartics, anti-inflammatory, antibacterial, antifungal and also as natural dyes

Pharmacological Activity

- Stimulant cathartics
 - Anthranols and anthrones elicit a more drastic action
 - The glycosides are hydrolyzed by the enzymes of the microflora
 - They act on the large bowel
 - Their effect may be delayed up to 6 hrs or longer
 - They increase the strength of peristaltic contractions but not the frequency
 - At larger doses, they cause colic pain
 - They are habit forming; they must not be prescribed for more than 15 days

Structure-Activity Relationship:

- Glycosylation is essential for activity
 - sugars are attached mainly at the C-8 or C-6 hydroxyl groups or as C-glycosides at the C-10 position
- Hydroxylation at C-1 and C-8 is essential for activity
- Oxidation level at C-9 and C-10 is important:
 - Highest level of oxidation (anthraquinones) have the lowest activity
 - Oxanthrones are less active than anthrones.
 - Complete reduction of C-9 and C-10 eliminates the activity.
- Substitution at C-3 have great impact on activity:
$$\text{CH}_2\text{OH} \quad > \quad \text{CH}_3 \quad > \quad \text{COOH}$$

Chemical Tests

Borntrager's test

- Requires free anthraquinone-type aglycones
- Aglycones produce pink, red or violet color with ammonium hydroxide solution or sodium bicarbonate solution
- Glycosides and reduced anthraquinone aglycones (anthrones, anthranols, oxanthrones and dianthrones) do not give positive result in this test
- Glycosides should be hydrolysed first by boiling the sample in 2N acid
- C-glycosides are hydrolysed by boiling the sample in 2N acid in the presence of ferric chloride
- Oxanthrones are treated with fuming nitric acid to convert them to anthraquinones

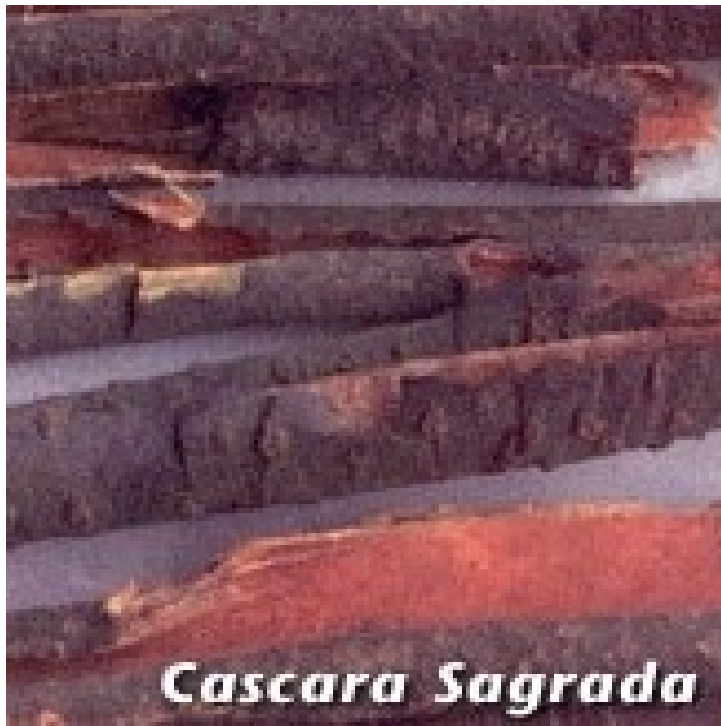
- **Schonteten's test (Florence test for anthranols)**

- Anthranols produce a strong green florescence in borax or other alkali solutions.

Examples of Crude Drugs Containing Anthraquinone Glycosides

Crude drug	Botanical source	Active constituents	Pharm act.: Med. use
Senna	Dried leaflets of <i>Cassia acutifolia</i> or <i>C. angustifolia</i> (fam. Fabaceae)	Sennosides A and B	Stimulant cathartics
Rhubarb	Dried rhizomes and roots <i>Rheum officinale</i> or other species of <i>Rheum</i> except <i>R. raponticum</i> (fam. Polygonaceae)	Sennosides, chrysophanol glycosides, aloe-emodine glycosides and emodin glycosides	Laxative
Cascara	Dried bark of <i>Rhumnus purshina</i> (fam. Rhamnaceae)	O-glycosides (10-20%), C-glycosides (80-90%) and some free anthraquinones: e.g. cascariosides	Laxative
Aloe	Dried juice of the leaves of various spp of <i>Aloe</i> like <i>Aloe barbadensis</i> (fam. Liliaceae)	Two fractions: an anthraquinone fraction called "aloin" and resins. barbaloin is the major constituent of the aloin.	Laxative

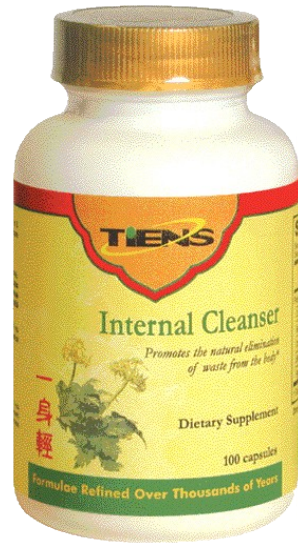
Cascara



Senna



Rhubarb



Aloe



Preparation of Aloes



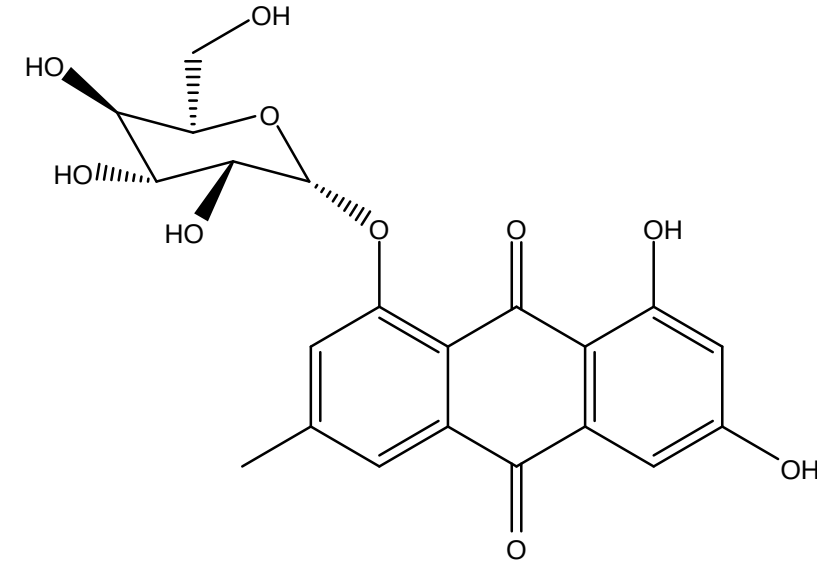
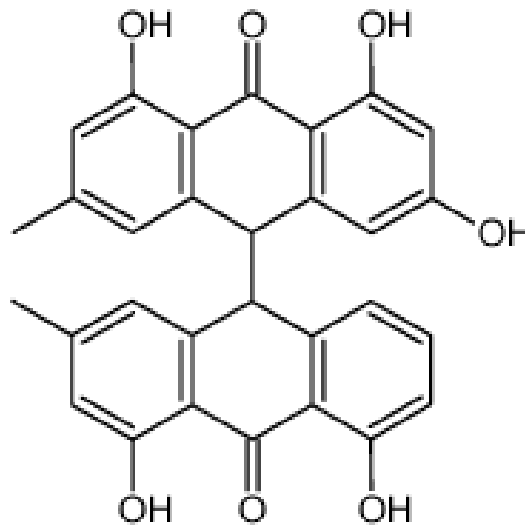
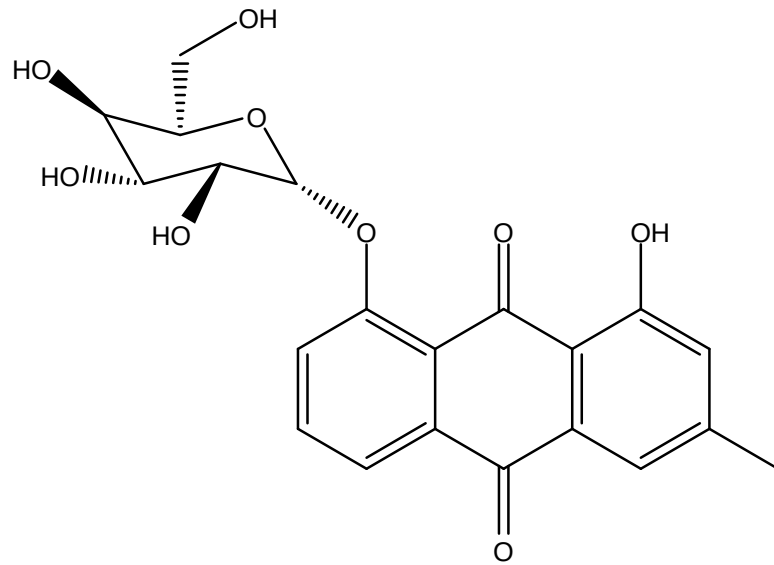


HYPOALLERGEN
+
KLINISCH GETESTET
Dermatest 10/2005
Note: Muito Bom

***Rumex abyssinicus* Jacq (Polygonaceae)**

- *Rumex abyssinicus* Jacq (Polygonaceae) has been used traditionally for treatment of hypertension, inflammatory and painful conditions in Ethiopia
- Reported to contain chrysophanol, physcion, emodin, palmidin C (emodin-chrysophanol bianthrone), chrysophanol-8-O- β -D-glucoside and emodin-8-O- β -D-glucoside



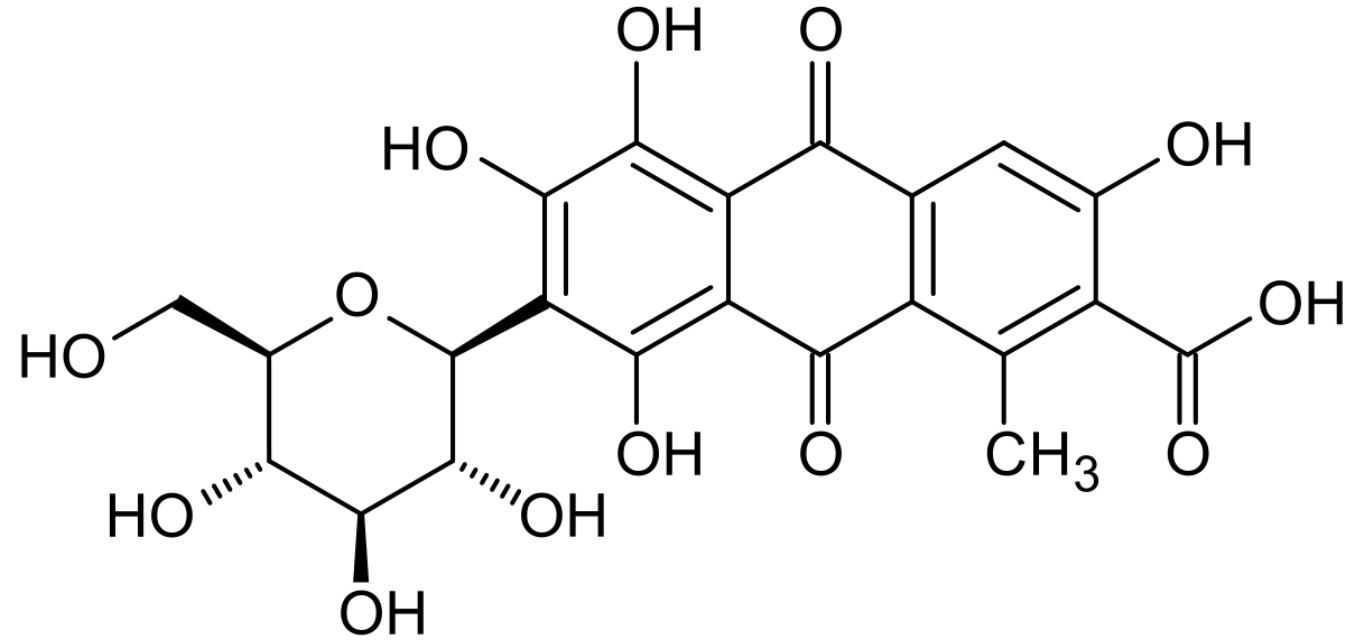


- chrysophanol-8-β-D-glucoside (on the left), Palmidin C (in the middle)
emodin-8-β-D-glucoside (on the right)

Carminic acid

- Cochineal (*Dactylopius coccus Costa*) infest on cactus pear (*Opuntia ficus-indica*)
- The female insect produces carminic acid to deter predators
- Carminic acid, typically 17-24% of dried insects' weight, can be extracted from the body and eggs, then mixed with aluminium or calcium salts to make carmine dye
- Carminic acid is a red dye used in food, cosmetic, pharmaceutical industries.





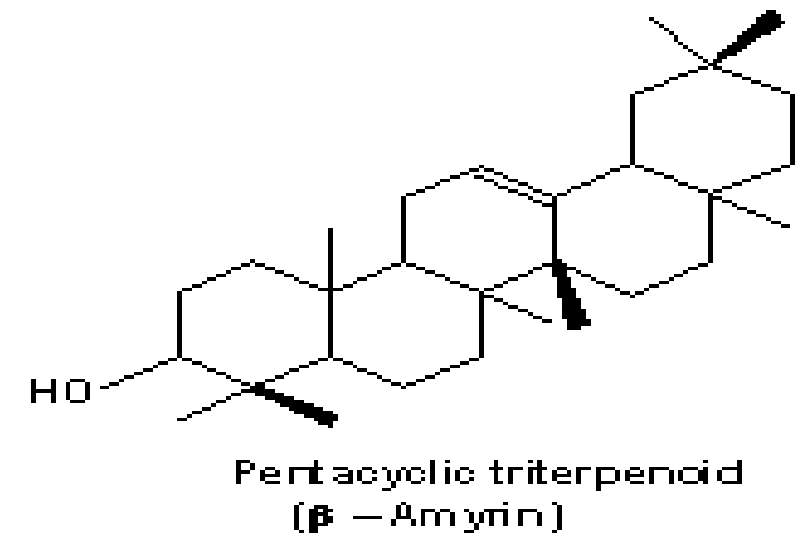
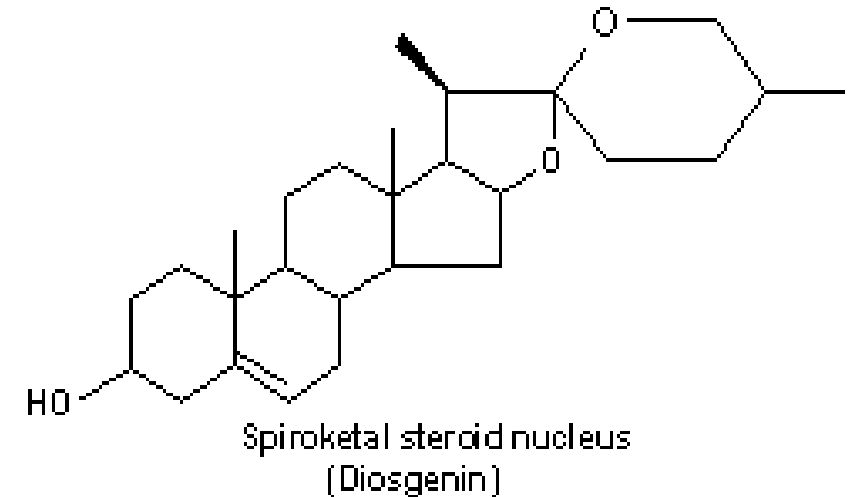
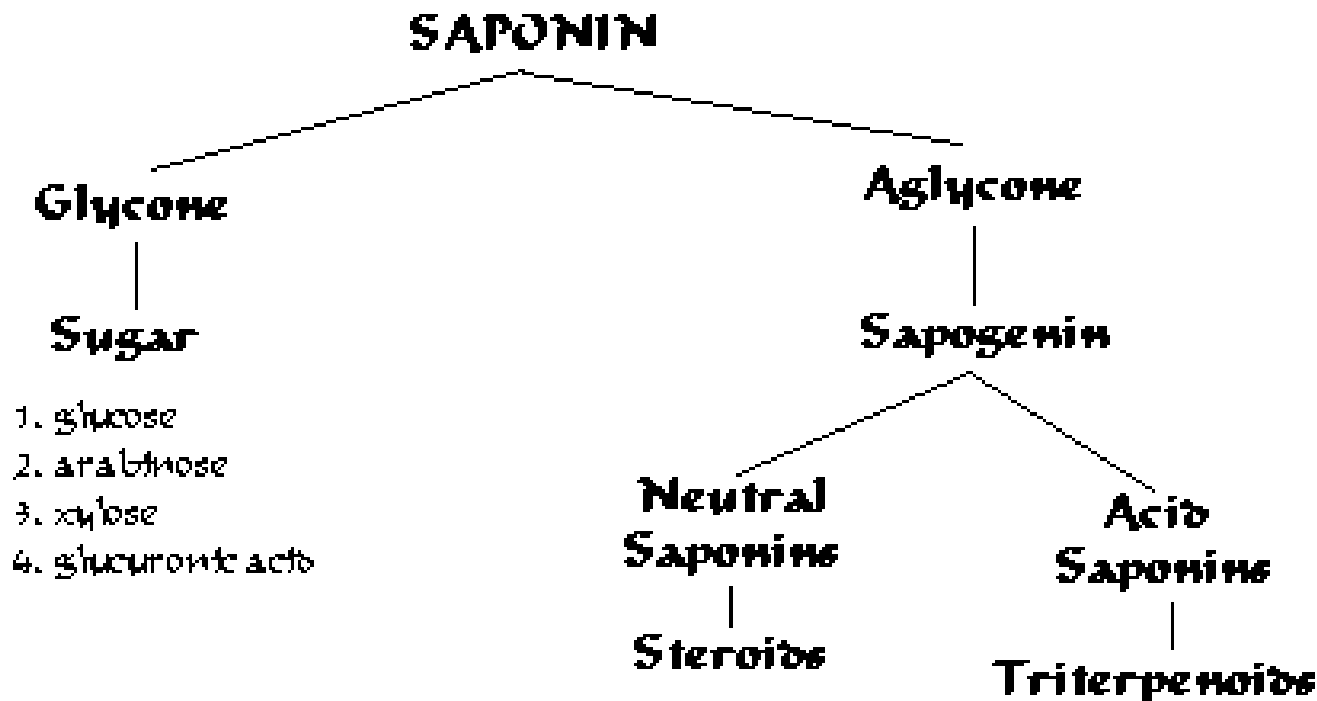
9 Foods Made With Crushed Bugs

Starbucks has announced they are reviewing alternative food dyes to cochineal (aka crushed bugs). But a Strawberry Frap isn't the only food you've eaten colored by crushed bugs.



2. SAPONIN GLYCOSIDES

- “saponin” is derived from the Latin word “sapo,” (Eng. “soap”)



Saponin Glycosides Cont...

- on dehydrogenation with selenium at 360°C
 - The steroid saponins give Diel's hydrocarbon
 - The triterpenoid gives naphthalene/picene
- have long been used in many parts of the world for their detergent properties

E.g. In South America bark of *Quillia saponaria*

In Europe root of *saponaria officinalis*

- Foam in aqueous solution by forming colloidal solution, which decreases surface tension
- Causes haemolysis of red blood cells hence toxic if injected to blood stream

2.1. Steroidal Saponins

- Less widely distributed in nature
- Common in:
 - Dioscoraceae (e.g. *discoria* spp),
 - Amaryllidaceae (e.g. *Agavae* spp)
 - Liliaceae (e.g. *yucca* and *trillium* spp)
- Derived from the mevalonic pathway (cholesterol)
- useful starting materials for the synthesis of steroidal drugs (hormones and contraceptives)
- Diosgenin, Hecogenin, Botogenin and their respective stereoisomers Yamogenin, Sisalgenin and Neobiogenin

2. 2. Triterpenoidal Saponins

- Abundant in many dicotyledonous families:
 - Caryophyllaceae
 - Sapindaceae
 - Polygalaceae
 - Sapotaceae

Uses

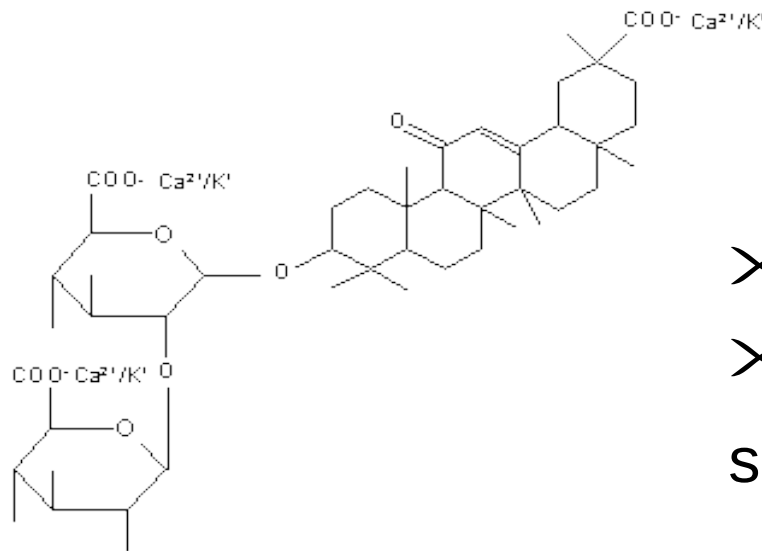
- Foaming beer (spirostane-type saponins from *yucca schidigera*)
- Cleaning industrial equipment and fine fabrics
- Powerful emulsifier of certain resin, fats and fixed oils
- Antifungal and antibacterial agents
- Lowering of blood cholesterol
- Inhibition of cancer cell growth

Examples of Crude Drugs Containing Saponin Glycosides

LICORICE (*Glycyrrhiza glabra*)

family *Leguminosae*

- Glycyrrhiza is the dried rhizome and roots of *Glycyrrhiza glabra*
- Contains a saponin glycoside called **Glycyrrhizin**

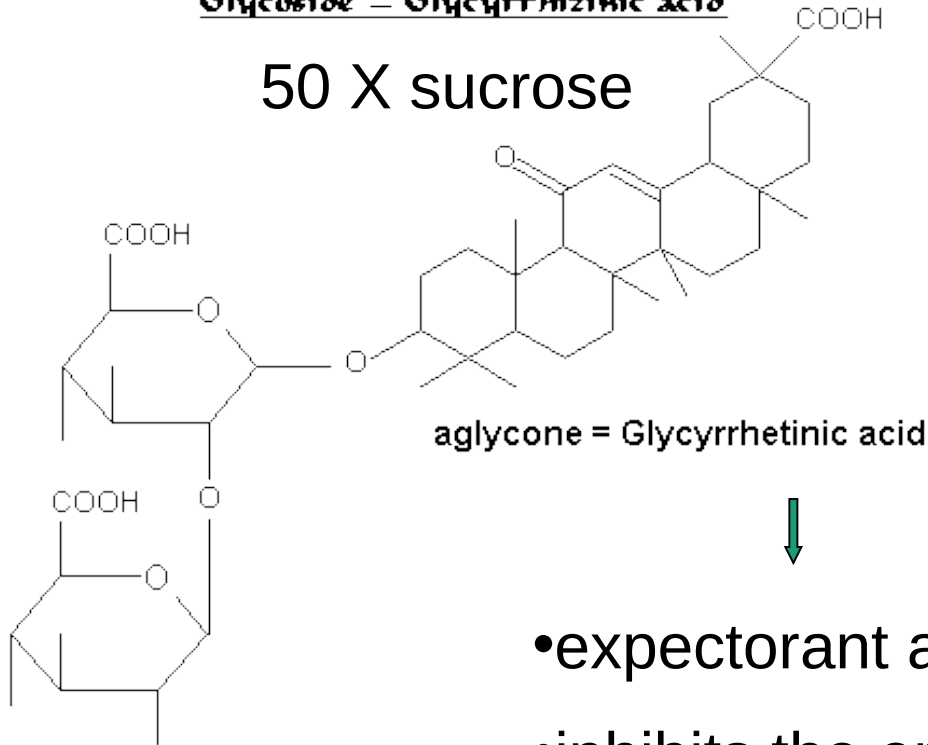


Glycyrrhizin = Ca²⁺ & K⁺ salts of Glycyrrhizinic acid

✂inhibits liver cell injury
✂It inhibits the growth of
several DNA and RNA viruses

Glycoside = Glycyrrhizinic acid

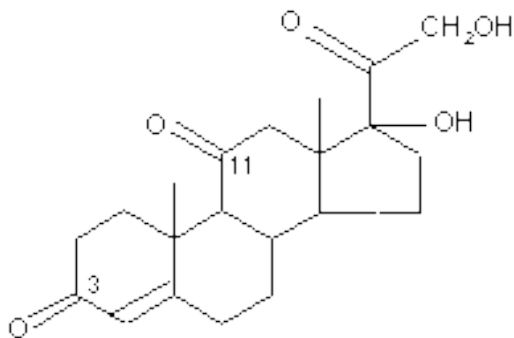
50 X sucrose



aglycone = Glycyrrhetic acid



2 molecules
of glycone = glucuronic acid



cortisone

- expectorant and antitussive properties
- inhibits the enzymes that metabolise the prostaglandins, PGE2 and PGF2alpha
- Potential for the treatment of peptic ulcer
- Anti-inflammatory

Contraindication

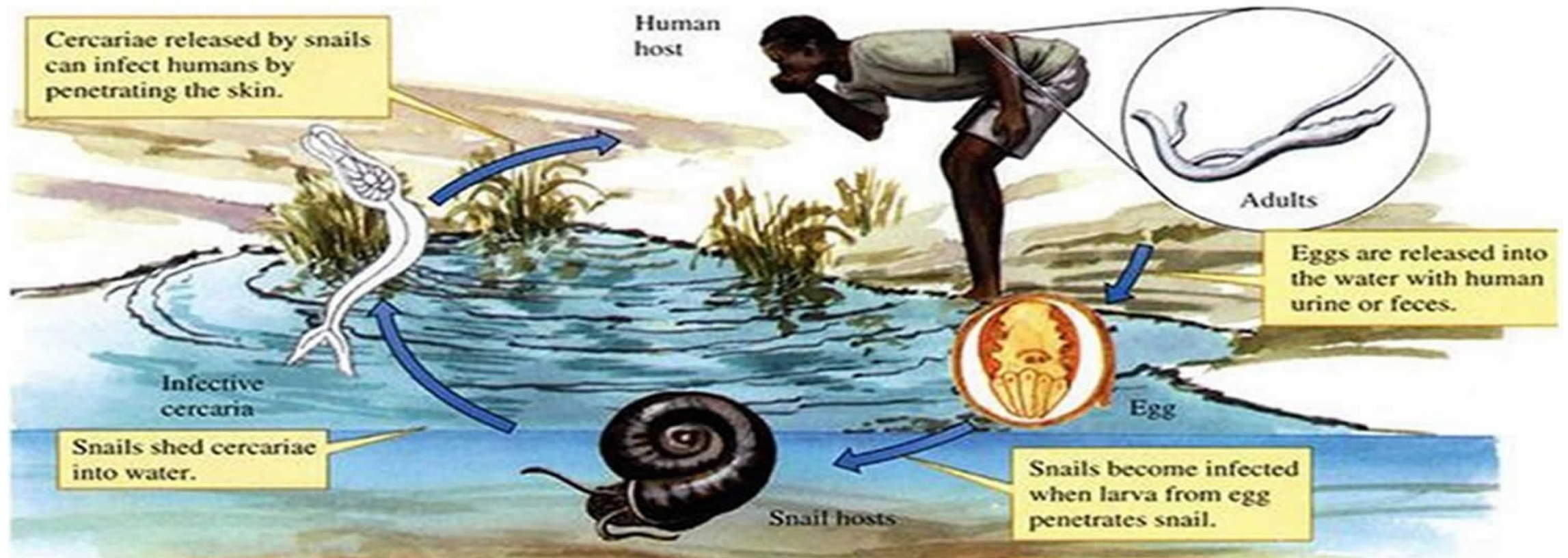
- Licorice should not be taken during pregnancy
- Licorice should not be given to patients with a known history of high blood pressure due to hypernatraemia
- Cardiac disorder

Licorice

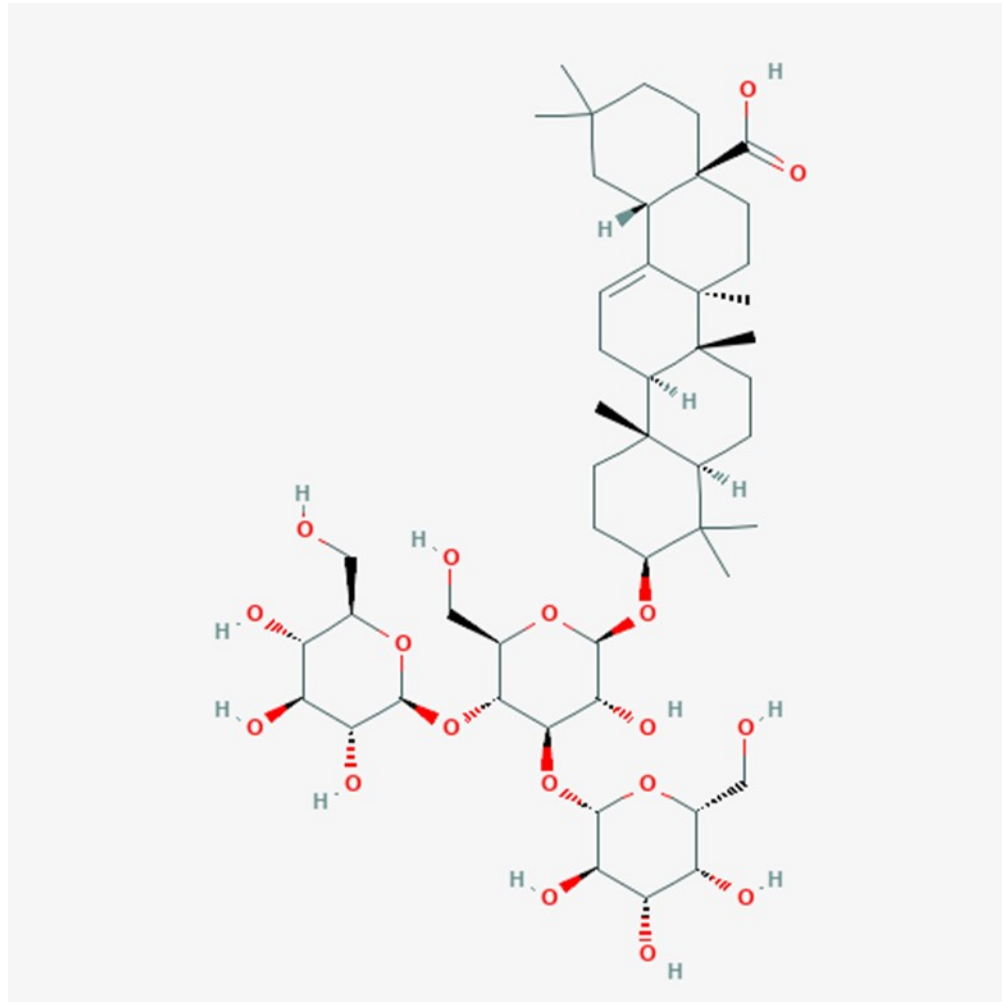


Phytolacca dodecandra





- “Endod” had properties of a molluscicide and a detergent.
- It kills snails, which vectors of schistosomiasis ("snail fever disease"), otherwise known as bilharzia.

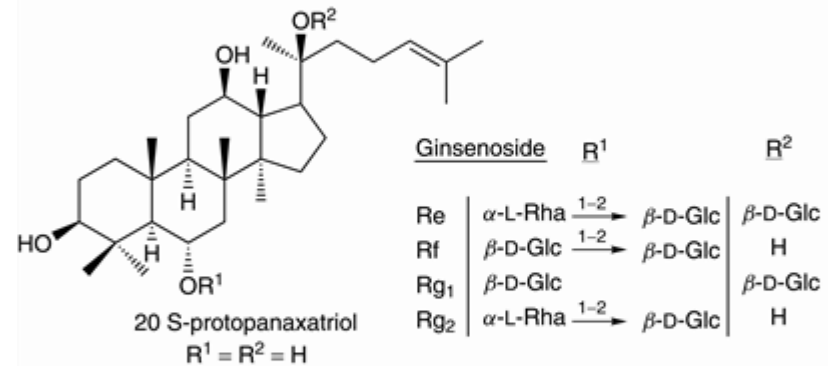
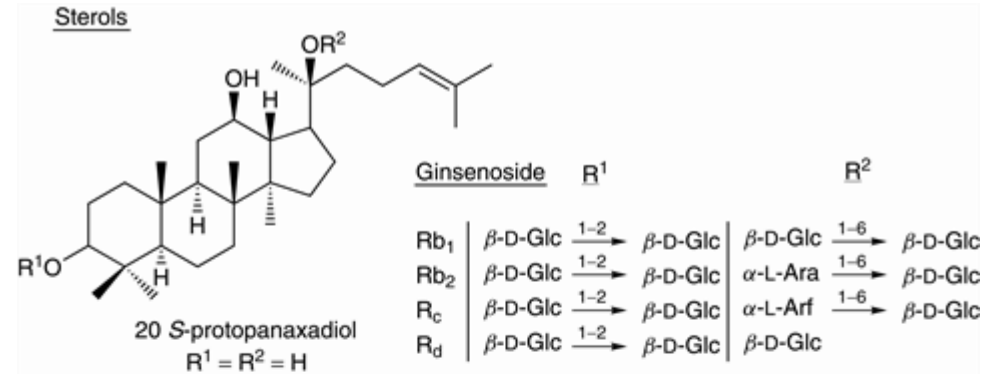


- Lemmatoxin isolated from *Phytolacca dodecandra*

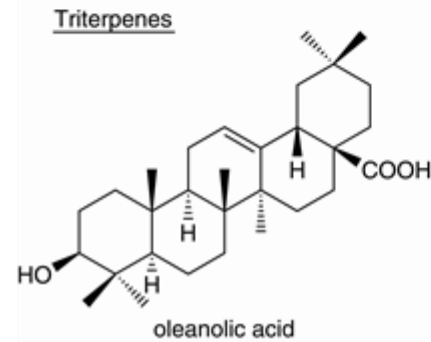
Ginseng

- For some 2000 years the roots of *Panax ginseng* C. A. Meyer (Araliaceae).
- Anaemia, diabetes, gastritis, sexual impotence
- Stamina, concentration, resistance to stress and to disease; in this sense the action of the drug is described as 'adaptogenic'.

Ginseng



Glc = glucose, Ara = arabinose, Arf = arabinofuranoside, Rha = rhamnose



Test for saponins

- Foaming index
- Hemolytic index

3. CARDIAC GLYCOSIDES

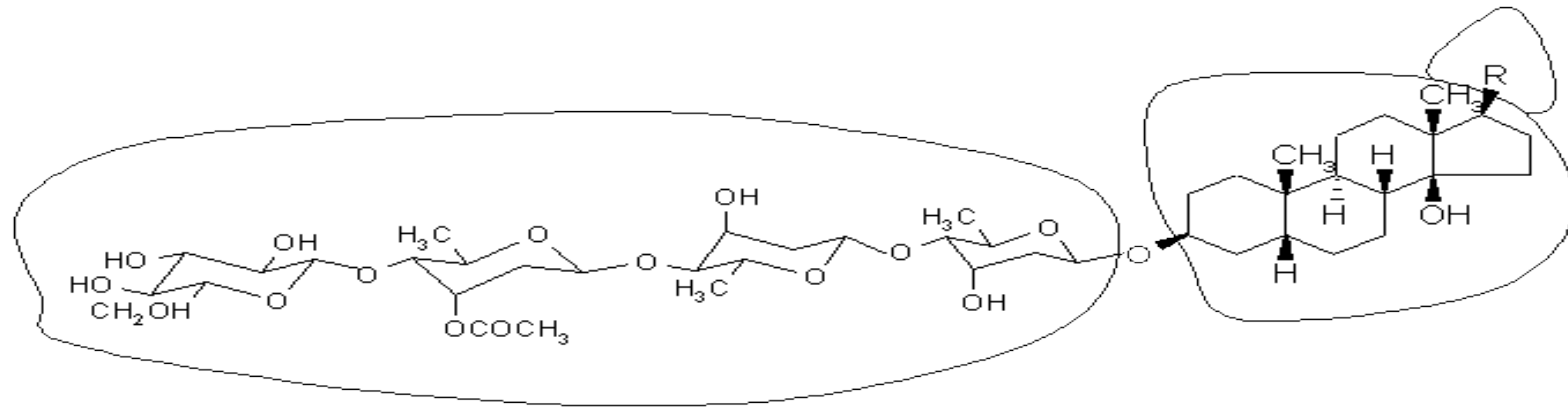
- Plant glycosides with specific action on heart
- Historical use:
 - to assassinate people, arrow poisons
- Historical sources:
 - South American toad skins, African plant extracts



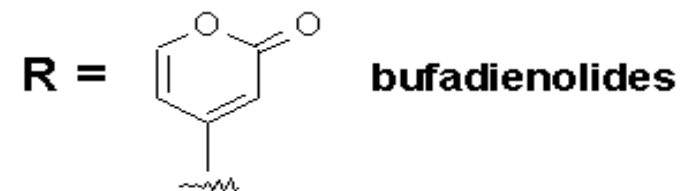
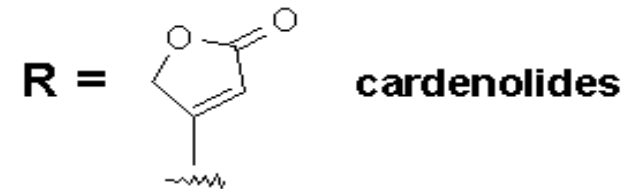
- Modern use:
 - to treat congestive heart failure (dropsy)
 - aglycone structure important for activity

Structure

- The R group at the 17-position defines the class of cardiac glycoside

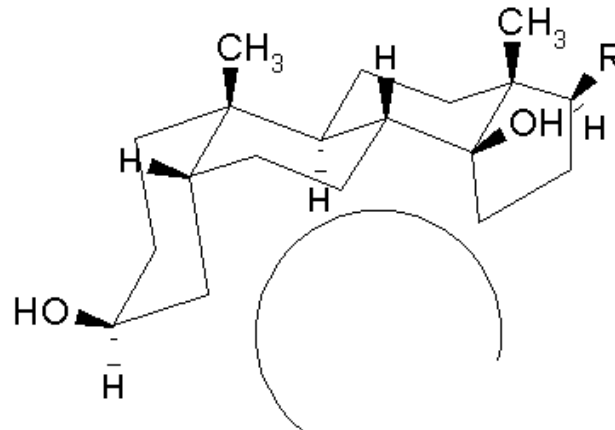


- Normally plant sources provide a 5-membered unsaturated lactone while animal sources give a 6-membered unsaturated lactone



Structure Cont ...

- Rings A/B and C/D are cis fused while rings B/C are trans fused.
- Such ring fusion gives the aglycone nucleus the characteristic 'U' shape as shown below



Sugars

- One to 4 sugars are found to be present in most cardiac glycosides attached to the 3 β -OH group
- The sugars most commonly used include L-rhamnose, D-glucose, D-digitoxose, D-digitalose, D-digginose, D-sarmentose, L-vallarose, and D-fructose

Structure - Activity Relationships

- The sugar moiety appears to be important only for the partitioning and kinetics of action
- The "backbone" U shape of the steroid nucleus appears to be very important
 - Structures with C/D *trans* fusion are inactive
 - Conversion to A/B *trans* system leads to a marked drop in activity
- The 14-OH group is now believed to be dispensable
- Lactones alone are not active
- The unsaturated 17-lactone plays an important role in receptor binding
 - Saturation of the lactone ring dramatically reduced the biological activity
- The lactone ring is not an absolutely requirement

Function

- Cardiac glycosides work by inhibiting the Na⁺/K⁺ pump
- This causes an increase in the level of sodium ions in the myocytes, which then leads to a rise in the level of calcium ions
- This inhibition increases the amount of Ca²⁺ ions available for contraction of the heart muscle
 - improves cardiac output and reduces distention of the heart.

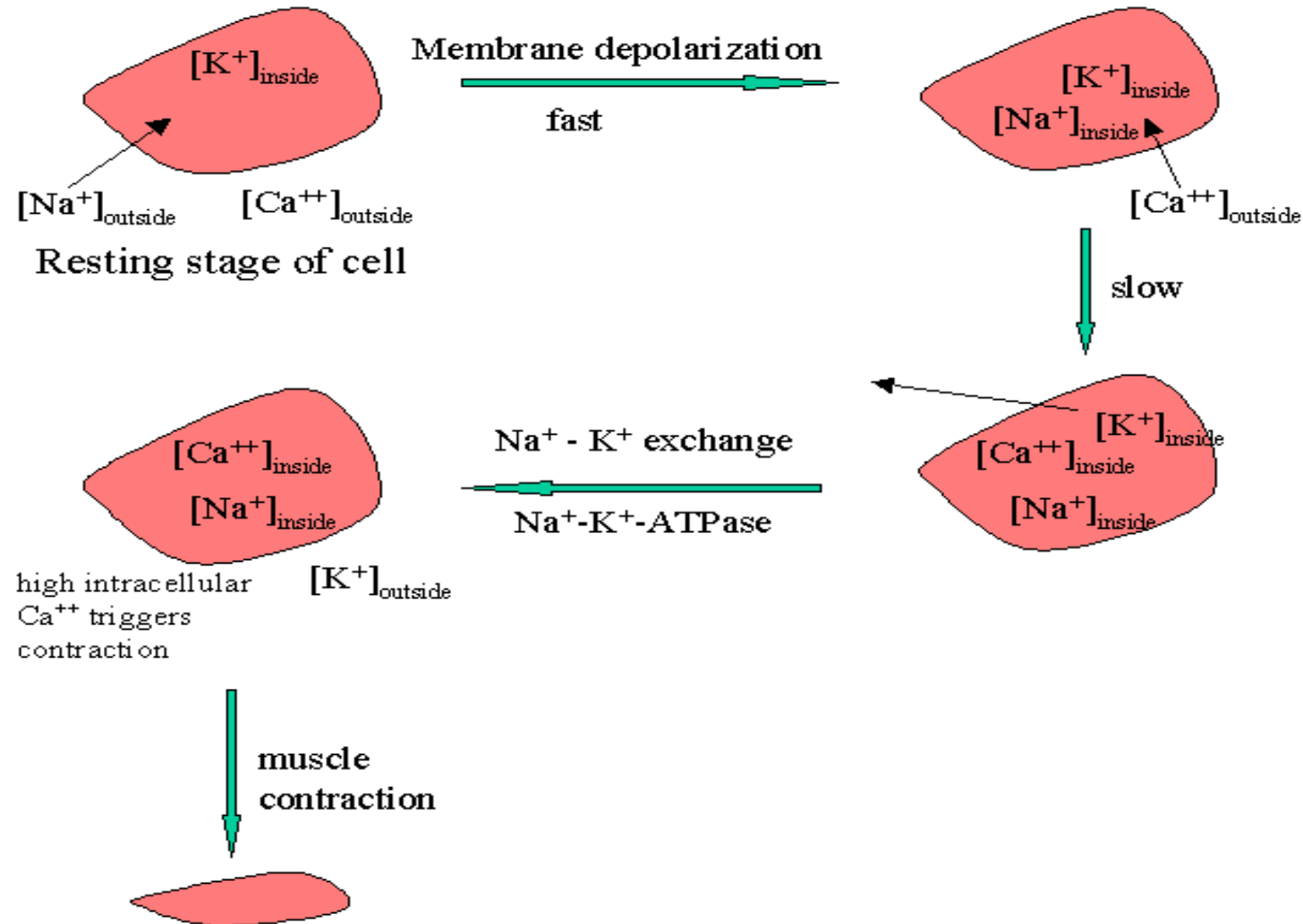
Mechanism

- They do this by stabilizing the E2-P transition state of the Na⁺/K⁺ pump.

The proposed mechanism is the following:

- inhibition of the Na⁺/K⁺ pump leads to increased Na⁺ levels
 - which in turn slows down the extrusion of Ca²⁺ via the Na⁺/Ca²⁺ exchange pump
 - increased amounts of Ca²⁺ are then stored in the sarcoplasmic reticulum and released by each action potential
- They have an antiarrhythmic effect by prolonging the refractory period of the AV node (Atrioventricular node)
 - reducing the number of impulses reaching the ventricles
 - cardiac output is restored but atrial fibrillation or atrial flutter are not abolished.

Biochemical Mechanism of Action



Sources

- Scrophulariaceae
 - *Digitalis purpurea* leaves (foxglove)
 - *Digitalis lanata* leaves – white flowers
- Apocyanaceae
 - *Strophanthus* vine seeds – Africa
- Liliceae
 - *Urginea* bulbs (squill) – Europe, India
 - *Convallaria* leaves (lily of the valley) – also produces a volatile oil perfume



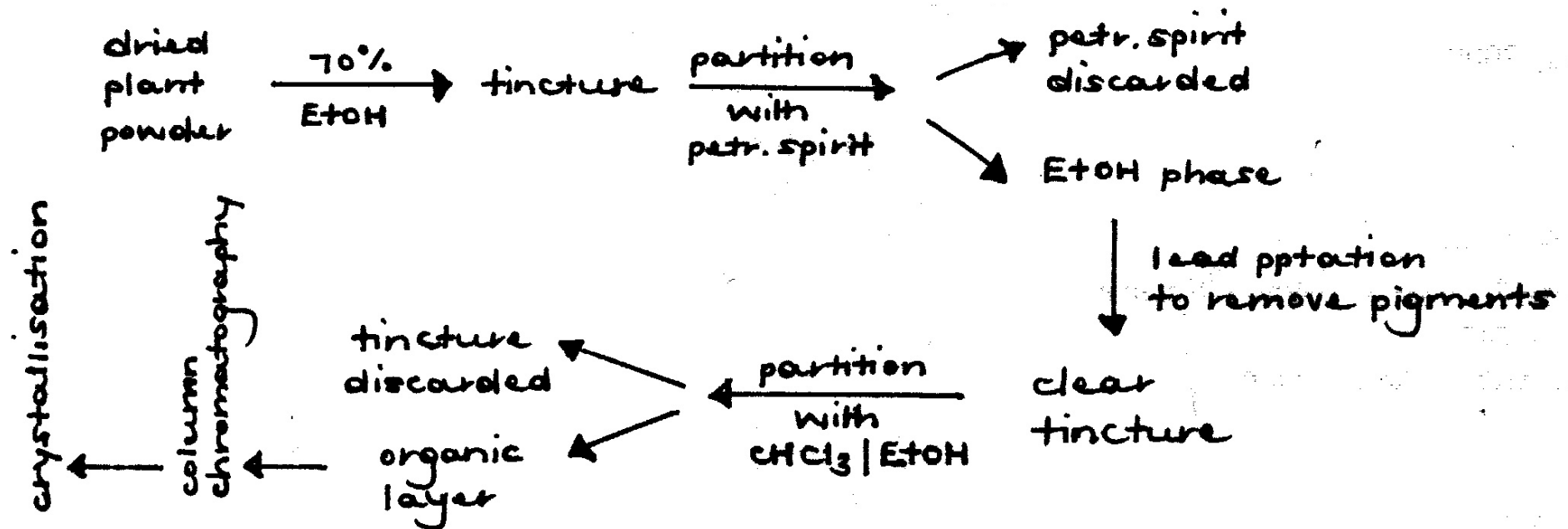
Digitalis

- Scrophulariaceae family
 - foxglove - biennial flowering plants
 - cases of poisoning rare
 - natural emetic if eaten in excess
 - *Digitalis purpurea* leaf – purple, British
 - -> Digitalis Tablets B.P.
 - -> Tincture of Digitalis B.P.
 - commercially grown Holland, E. Europe
 - NB no extraction for these products
 - *Digitalis lanata* leaf – white, Mediterranean
 - used for manufacture of pure glycosides
 - ie digoxin, lanatoside C
 - commercially produced Holland, Equador, USA



Extraction

- large molecular weight molecules with sugars -> polar
 - soluble in water and alcohol



- expensive - long process, solvents

Detection

Colour reactions due to aglycone moiety:

- These are characteristic for the 5-membered lactone ring of cardenolides:
 - 1- Kedde's test: card. + Kedde's reagent (3,5-dinitrobenzoic acid + NaOH) → violet colour.
 - 2- Baljet's test: card. + Baljet's reagent (picric acid + NaOH) → orange or red.

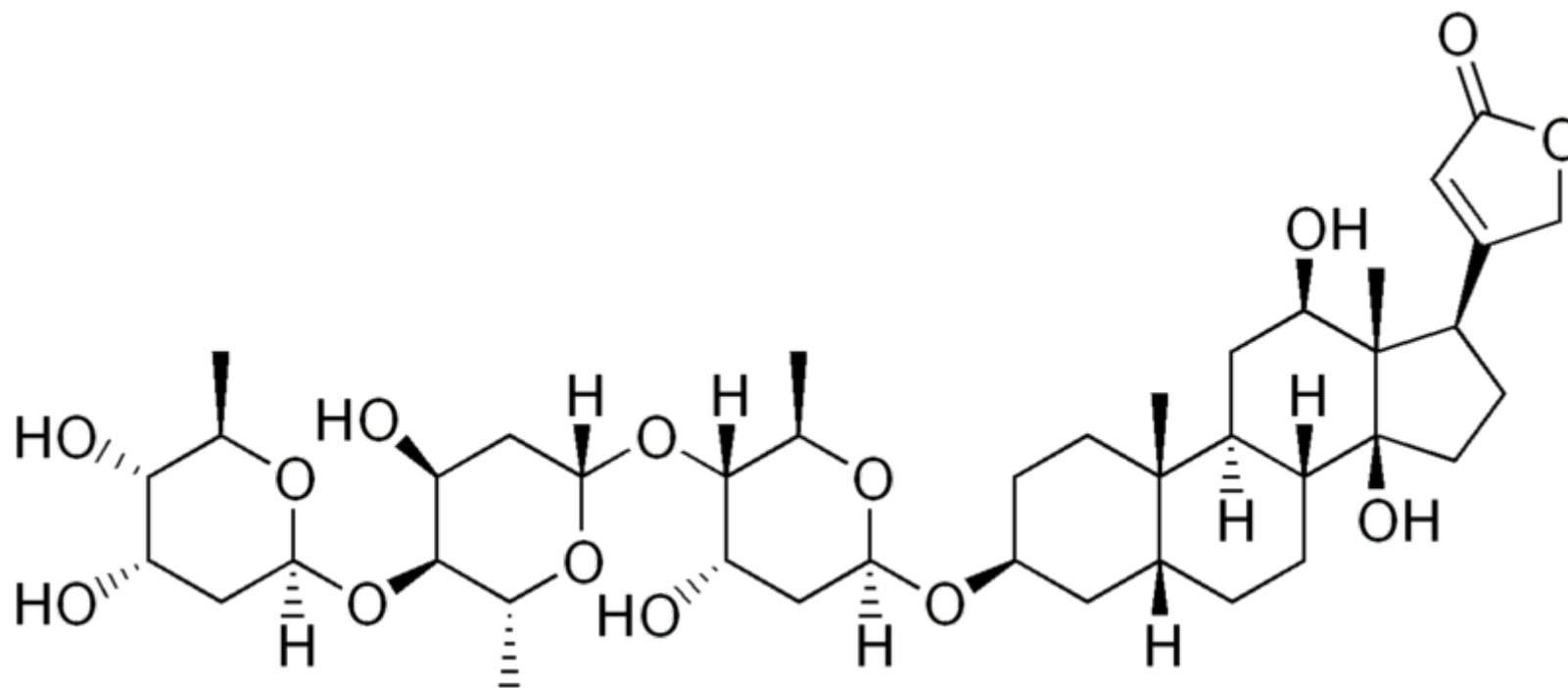
Detection Cont...

- **Reactions due to the steroidal nucleus:**
 - Liebermann's test: card. in glacial acetic acid + $\text{H}_2\text{SO}_4 \rightarrow$ red, violet, blue to green
- **Reactions due to the presence of 2-deoxy sugar unit in the sugar moiety:**
 - **Keller-Killiani's test:** card. in glacial acetic acid containing traces of FeCl_3 , conc. H_2SO_4 on the wall of the test tube $\rightarrow$ blue ring is formed in between the two layers

Digoxin

- is widely used in the treatment of various heart conditions, namely atrial fibrillation, atrial flutter and sometimes heart failure that cannot be controlled by other medication
- preparations are commonly marketed under the trade names **Lanoxin**, **Digitex**, and **Lanoxicaps**
- It is also available as a 0.05 mg/mL oral solution and 0.25 mg/mL or 0.5 mg/mL injectible solution
- It is marketed by GlaxoSmithKline

STRUCTURE

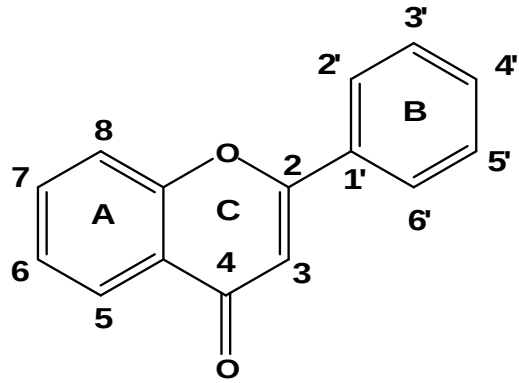


4. FLAVONOID GLYCOSIDES

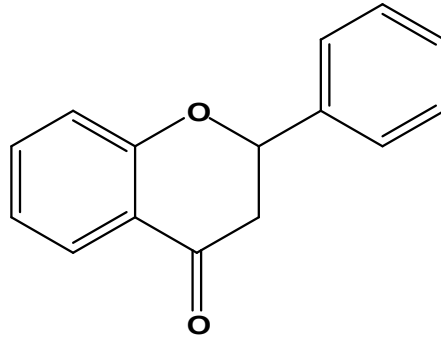
- the aglycone has a $C_6C_3-C_6$ nucleus
- are usually substituted by hydroxyl groups
- are responsible for the color of flowers, fruits and leaves
- Occur:
 - mainly as O-glycosides
 - Sometimes as C-glycosides and free aglycones



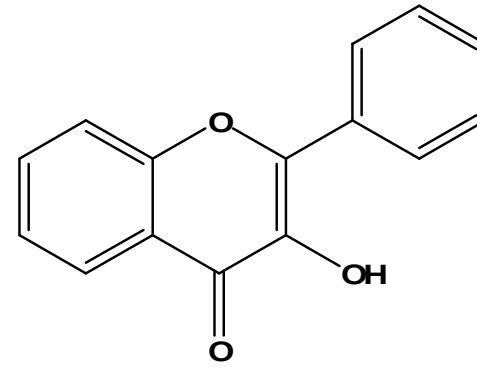
Structures



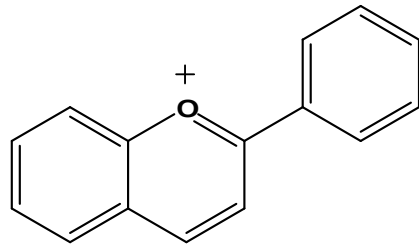
Flavone



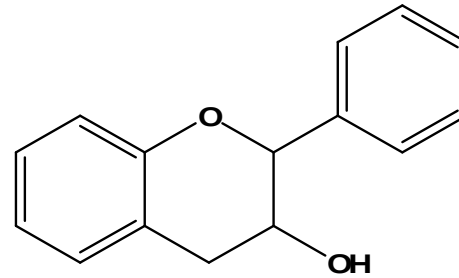
Flavanone



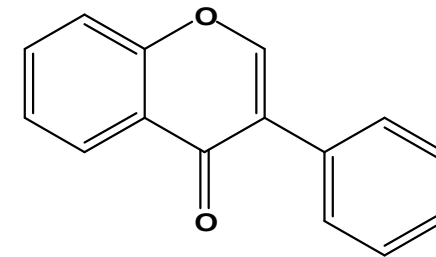
Flavonols



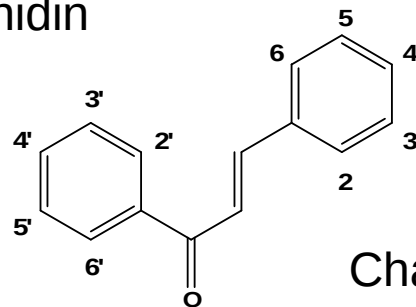
Anthocyanidin



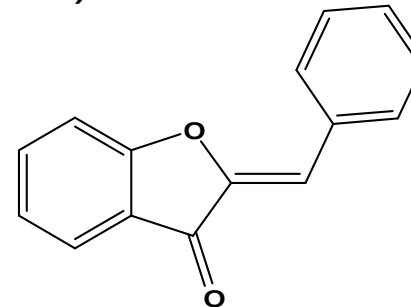
Catechin (flavan-3-ol)



Isoflavanoid



Chalcone



Aurones

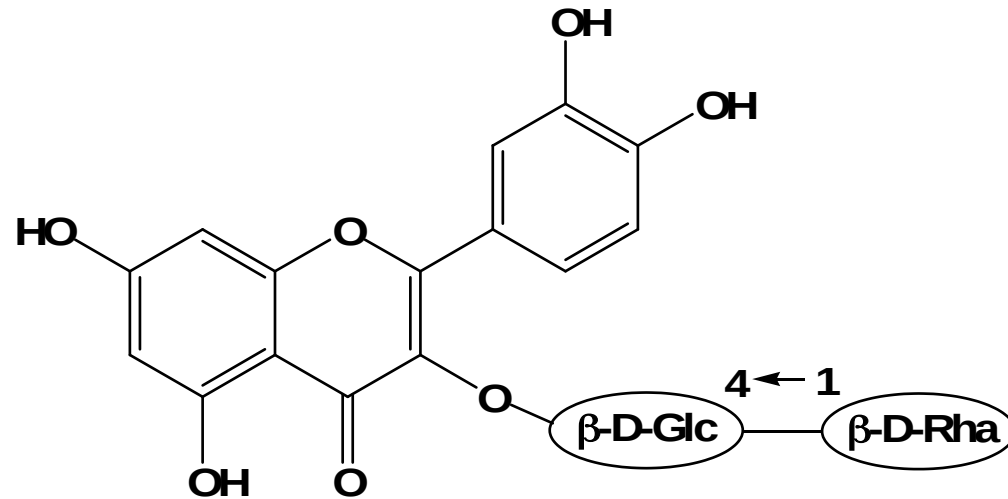
Flavonoids Cont...

- Many have yellow color
- Anthocyanins have red, blue or purple color
- Most absorb in the UV region
- exhibit various pharmacological activities including anti-inflammatory (and analgesic) properties
 - anti-allergic effects
 - antithrombotic, vasoprotective properties
 - decrease capillary fragility
 - phlebitis – changes in vessel walls in extremities
 - > plasma leakage -> oedema
 - mainly due to high oestrogen, sometimes in males
 - tumour inhibition promotion
 - protective for gastric mucosa
 - antioxidant
 - anti-inflammatory, anti-HIV

Some Medicinally Important Flavonoids

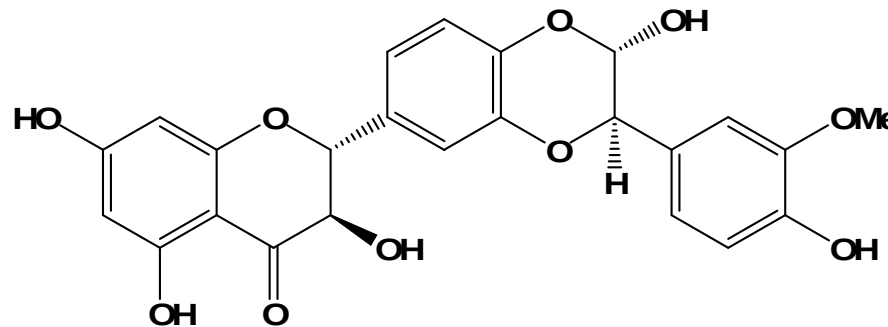
- Rutin

- Used as for the treatment of capillary bleeding and increased capillary fragility



- **Silybin**

- Silybin is the major active compound isolated from silymarin
 - a hepato-protective Flavonoidal fraction obtained from seed/fruits of ***Silybum marianum*** (Asteraceae) (Milk Thistle, St. Mary Thistle)
 - It is a mixture of mainly three flavonolignans, viz, silybin of, silidianin and silychristine.
- Silybin and also silymarin are used for wide range of liver problems



- Silymarin is included in the pharmacopoeia of many countries under the trademark Legalon or Hepatron
- is often used as supportive therapy in food poisoning due to fungi and in chronic liver disorders, such as steatosis and alcohol-related liver disease

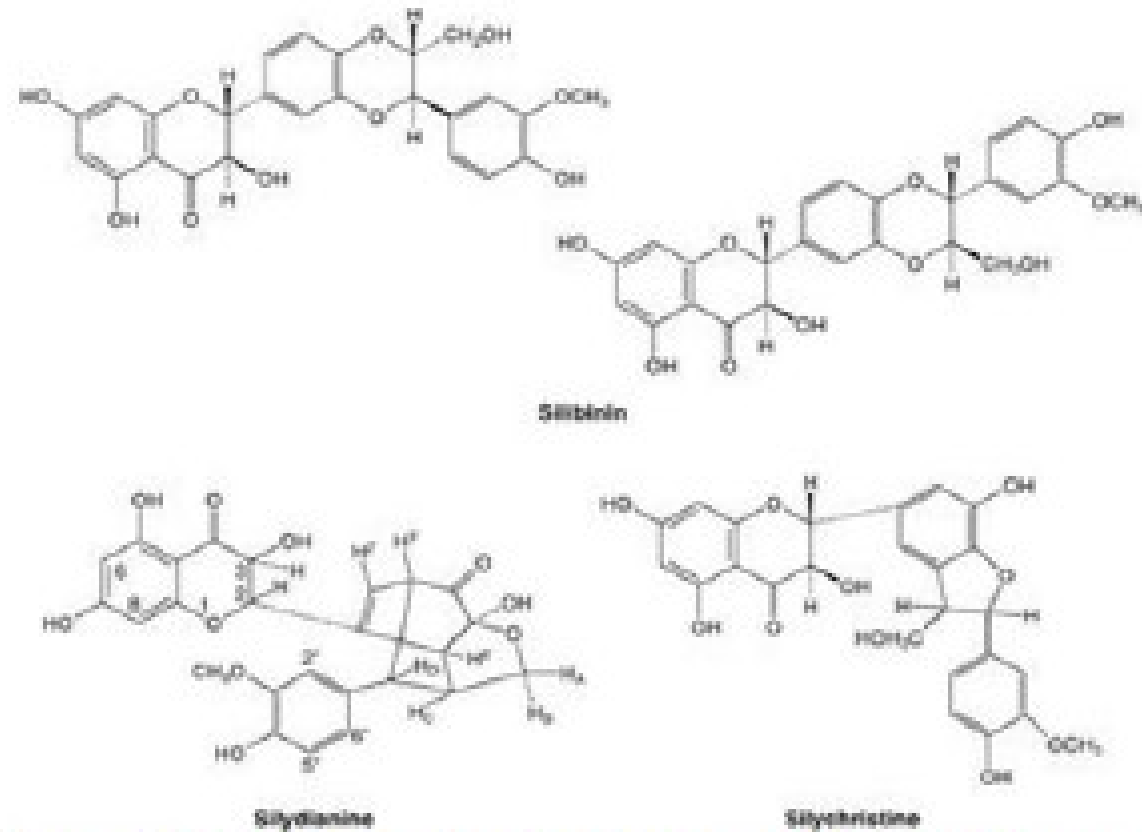


Figure 1: The three structural components of silymarin: silibinin, silydianine, and silychristine



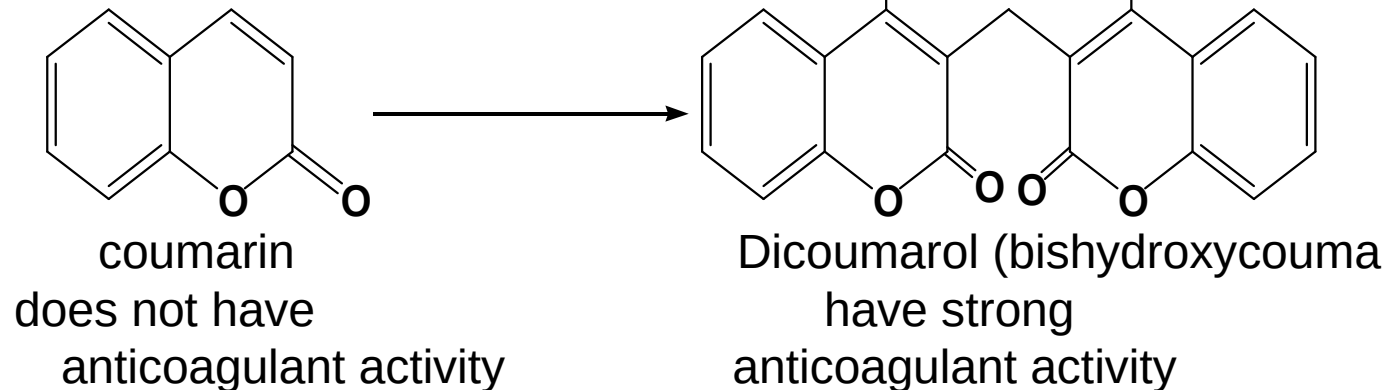
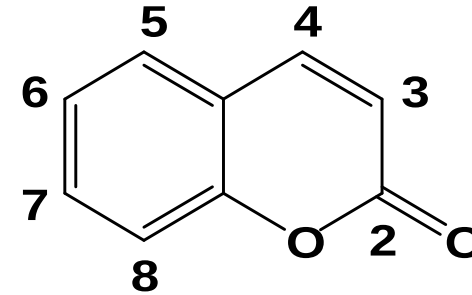
Isolation

- easy
- water and alcohol soluble
- give brightly coloured solutions
- crystallise easily
- may give colour reactions eg
 - MgCl_2 -> violet -> orange
 - alkali KOH -> orange
- easy to detect



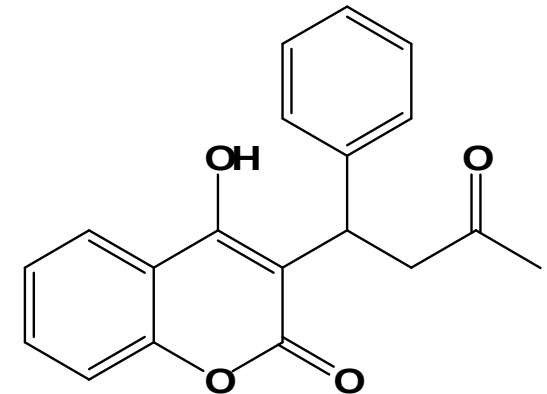
5. COUMARIN GLYCOSIDES (LACTONE GLYCOSIDES)

- some have anticoagulant activity
- They are widely distributed in nature
 - Leguminosae
 - Rubiaceae
 - Umbelliferae
 - Solanaceae
- They are converted to derivatives with such activity, by microbial action (e.g. dicoumarol)



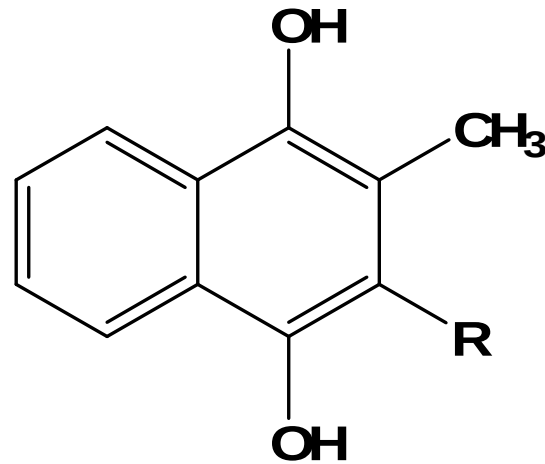
Coumarin Glycosides Cont...

- The structural requirements:
 - 4-hydroxy group
 - A substituent at C3.
- Dicoumarol is very potent anticoagulant which can not be used for treatment purpose
- It was used as a lead compound for the synthesis of coumarin based oral anticoagulants such as **warfarin**



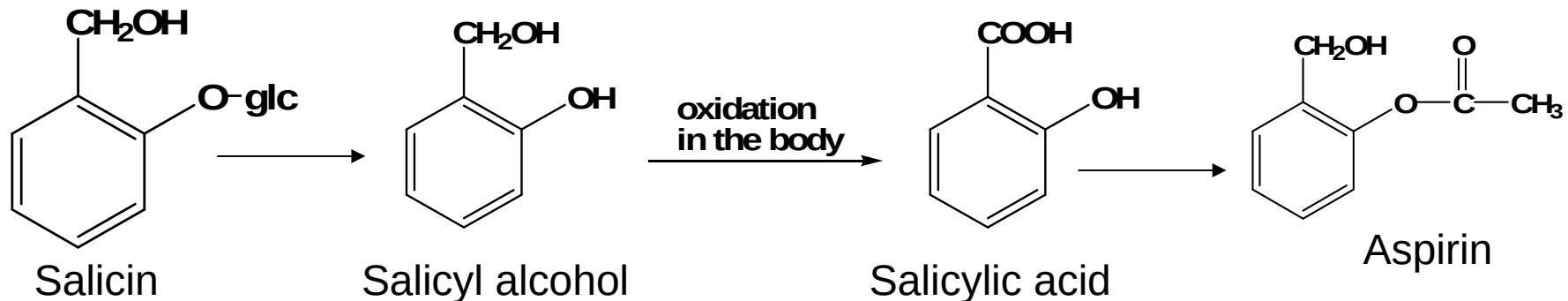
Mechanism of Action

- are competitive inhibitors of vitamin K in the biosyntheses of functional thrombine (vitamin K epoxide reductase).
- have structural resemblance to vitamin K



6. ALCOHOL GLYCOSIDES

- A notable example is salicin
 - is isolated from willow bark (*Salix fragilis* or *S. purpurea*: fam. Salicaceae).
- Traditionally willow bark has been and still is used
 - for fever and flu-like symptoms
 - as an analgesic (for headaches and toothaches)
 - orally or topically for symptomatic relief of minor pains of the joint and rheumatoid disorders



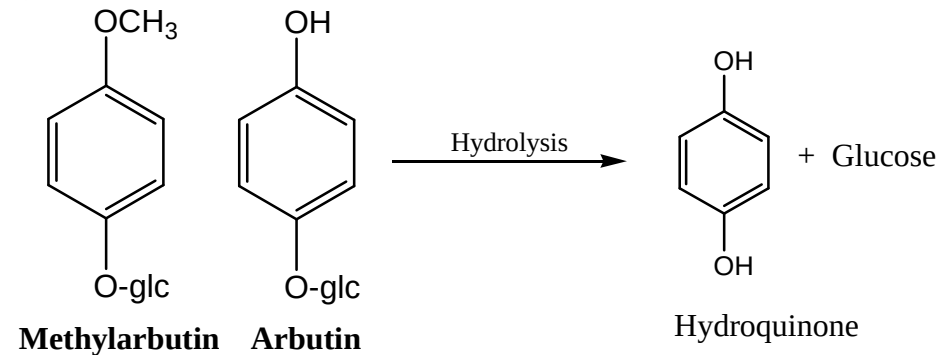


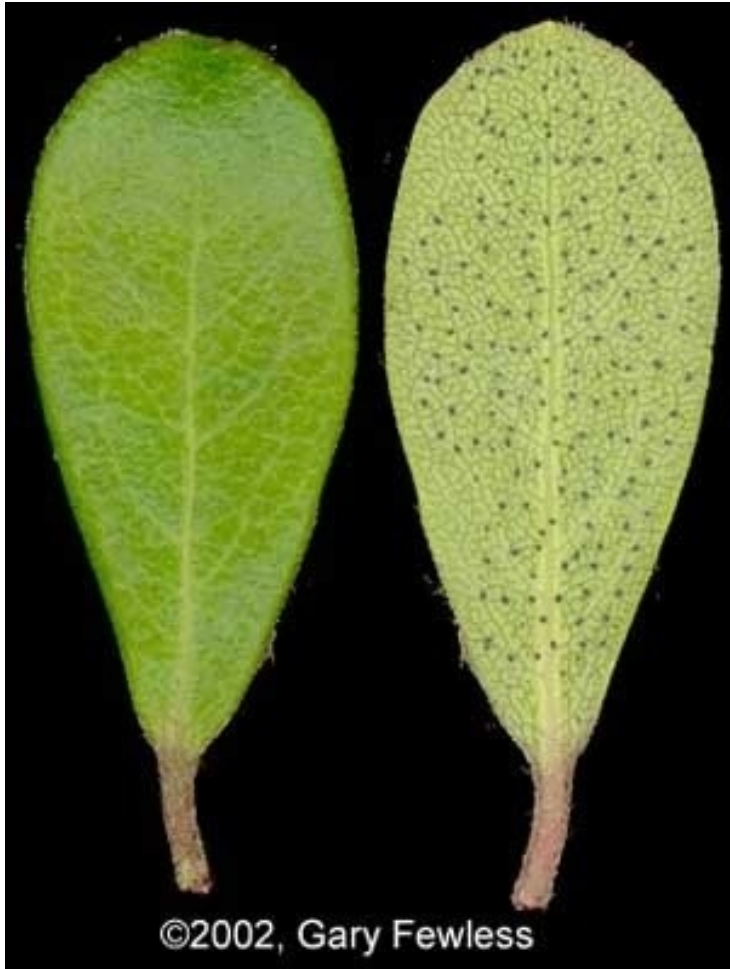
Arbutin & Methyl Arbutin

Source: *Uva ursi* (Bearberry leaves).

➤ **Nature:** Primary Phenolic glycoside (monoside)

➤ **Uses:** Diuretic- Bactericidal

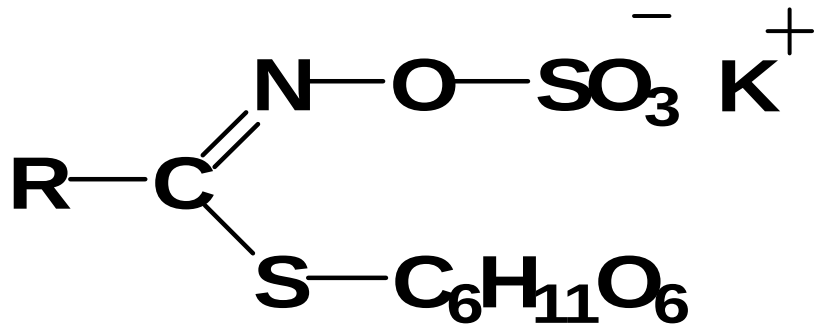




7. ISOTHIOCYANATE GLYCOSIDES (GLUCOSINOLATES OR THIOGLUCOSIDES)

General formula:

- They are S-glycosides widely distributed in family Cruciferae
- Can be separated on ion exchange resin

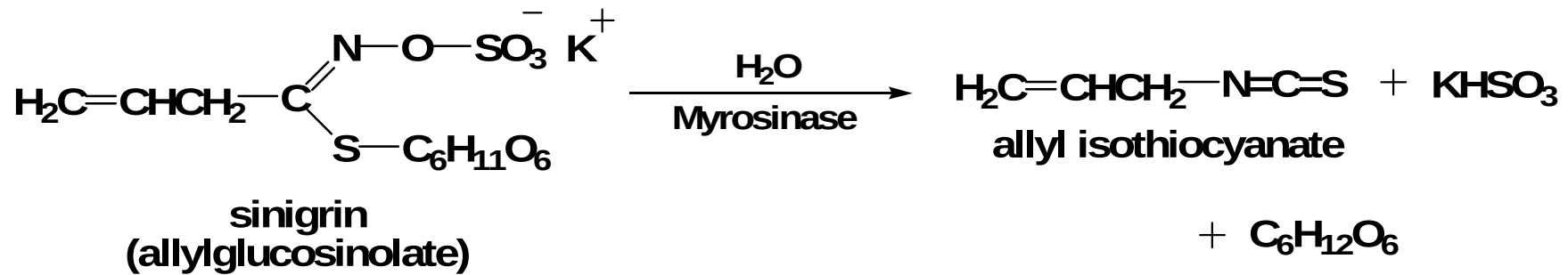


Sinigrin: In seeds of *Brassica nigra* (black mustard)

Sinalbin: In Seeds of *Brassica alba* (white mustard)

Sinigrin

- Occurs in the seeds of black mustard (*Brassica nigra*: fam. Brassicaceae) or Indian mustard (*B. juncea*: fam. Brassicaceae)



- Allyl isothiocyanate is strong-smelling volatile compound
 - Responsible for the astringent odor and flavor of mustard

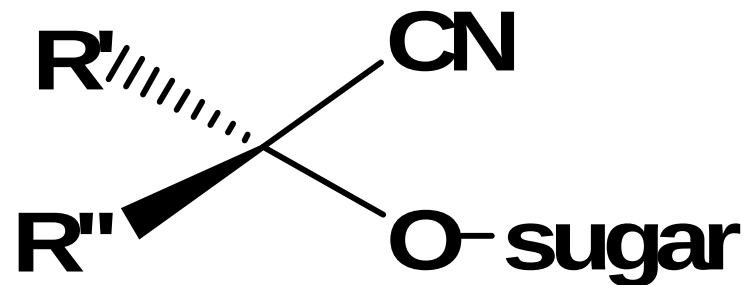


Uses

- Uses of allyl isothiocyanate
 - as condiment
 - is used as counter irritant (rubefacient)
 - used in the form of plasters

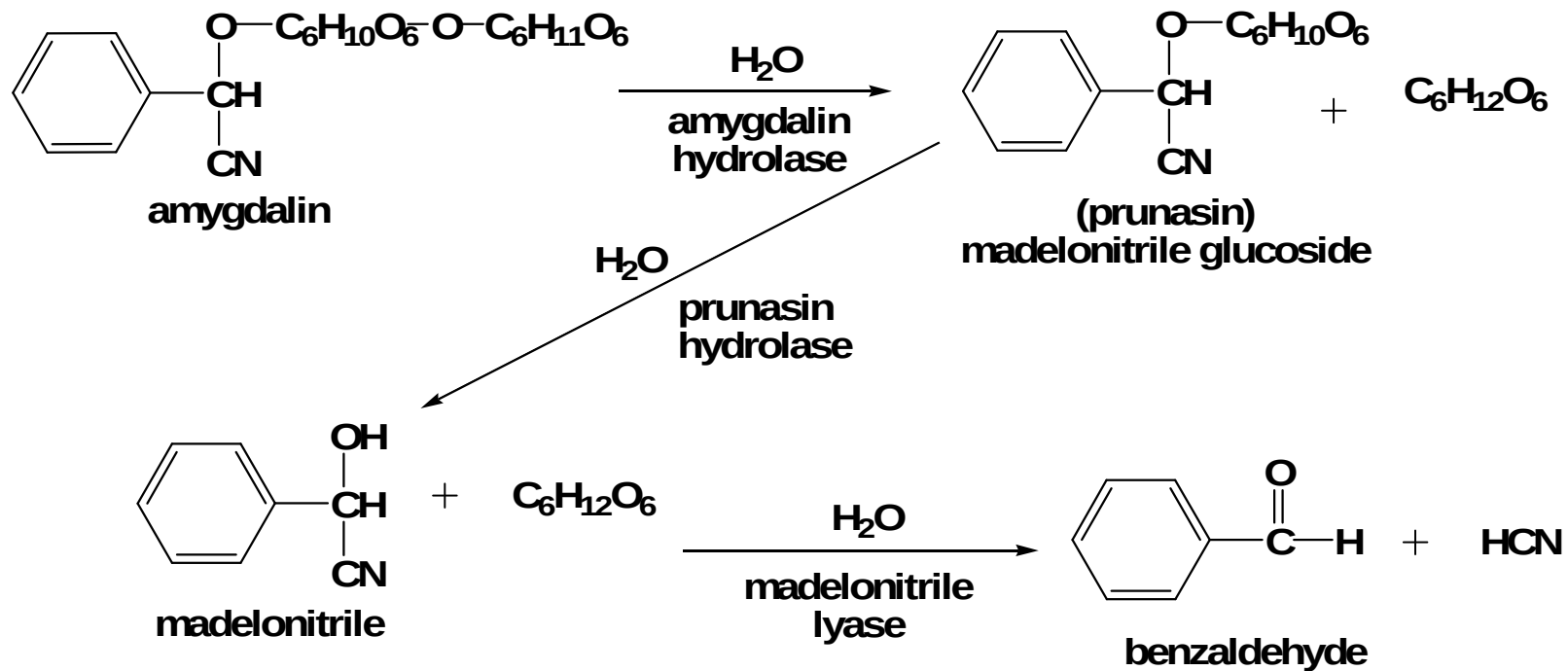
8. CYANOGENIC GLYCOSIDES

- Produce HCN upon complete hydrolysis which is toxic
 - Oral dose of 50 mg HCN can be fatal. How???
- All cyanogenic glycosides are of 2-hydroxynitriles



Occurrence and Distribution

- Over 2500 plant species
- Family Rosaceae, Fabaceae, Poaceae etc.
- Examples:
 - apricot , bitter almond, apple, plum, wild cherry (bark)
 - cassava (*Manihot esculenta*: Euphorbiaceae)
 - linseed (*Linum usitatissimum*: Linaceae)

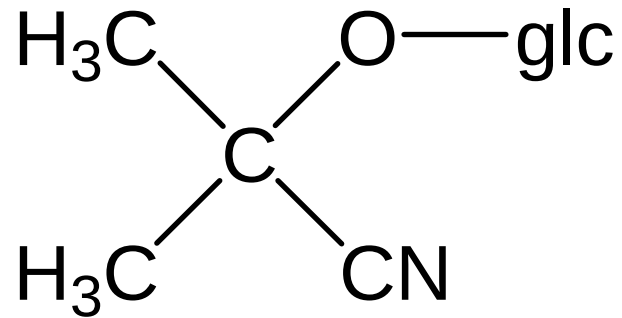


- Plants containing amygdaline are commonly employed for the preparation of benzaldehyde which is used as flavor

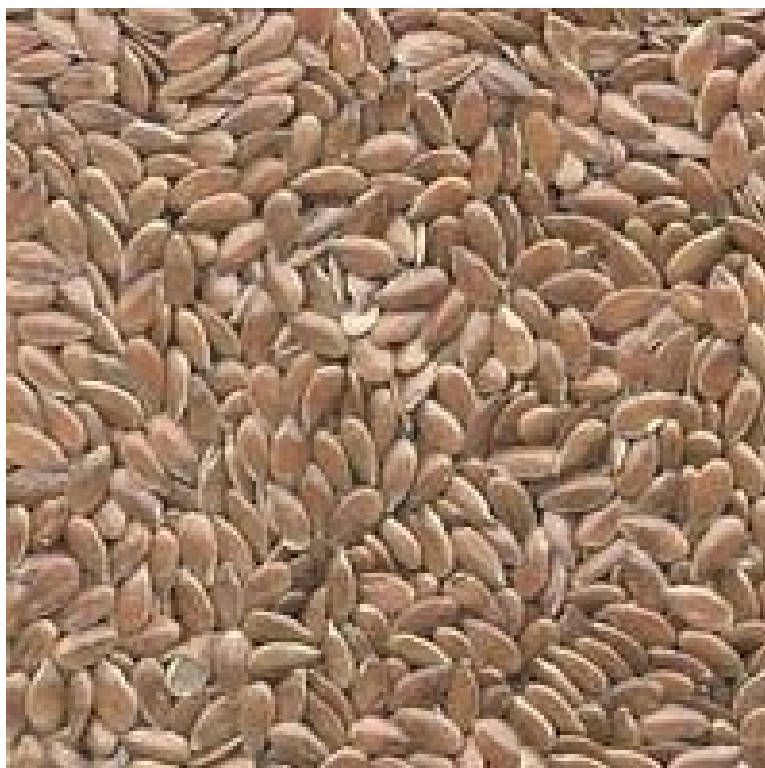
Linamarin

Source: Linseed

- **Structures:** It is the glycosidic derivative of the cyanohydrin of acetone.



- **Uses:**
 - Linamarin has a molluscicidal activity.
 - Cyanogenic glycosides have role in cancer treatment



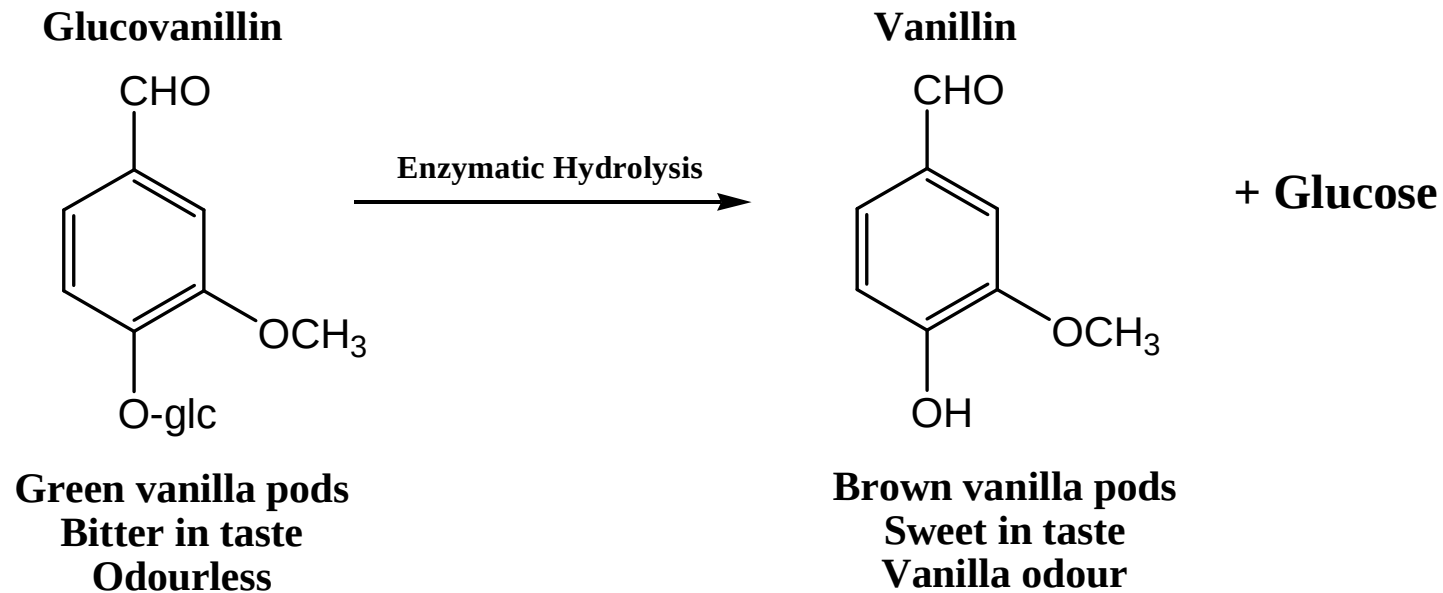
Test for Cyanogenic Glycosides

- Reduce plant material to small pieces and moisten with water
- Incubate at temp. less than 45 °C for 30 to 60 mins with the neck of the flask stoppered and have suspended sodium picrate paper
- The paper will turn brick red due to the release of HCN gas

9. ALDEHYDIC GLYCOSIDES

Glucovanillin

- Source: Vanilla pods.
- Uses: Flavouring agent- Spray reagent





Commercial Preparation of Vanillin

